

EXPERIMENT 6

ROLL NO: 240701230

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USER INTERFACE DESIGN

Aim

To design and simulate the stages of User Interface development using the Rapid Application Development (RAD) model by creating a food delivery mobile interface prototype using Balsamiq.

Tools Used

- Balsamiq (for wireframing and UI design)
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Introduction to RAD Model

1. Rapid Application Development (RAD) is a software development approach that focuses on quick design and prototype creation with continuous improvements based on feedback. It reduces development time by emphasizing working models instead of lengthy planning.
 2. The RAD model includes the following phases:
 3. Requirement Planning
 4. User Design
 5. Construction
 6. Testing and Deployment
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RAD Lifecycle Applied to the Project

1. Requirement Planning

The application developed is a Food Delivery Mobile App.

Objective:

To create an easy-to-use interface that allows users to search for food, explore restaurants, and place orders quickly.

Target Users:

Online food customers

Students and working professionals

Key Features:

Search food or restaurants

View restaurant details

Add food items to cart

Place order

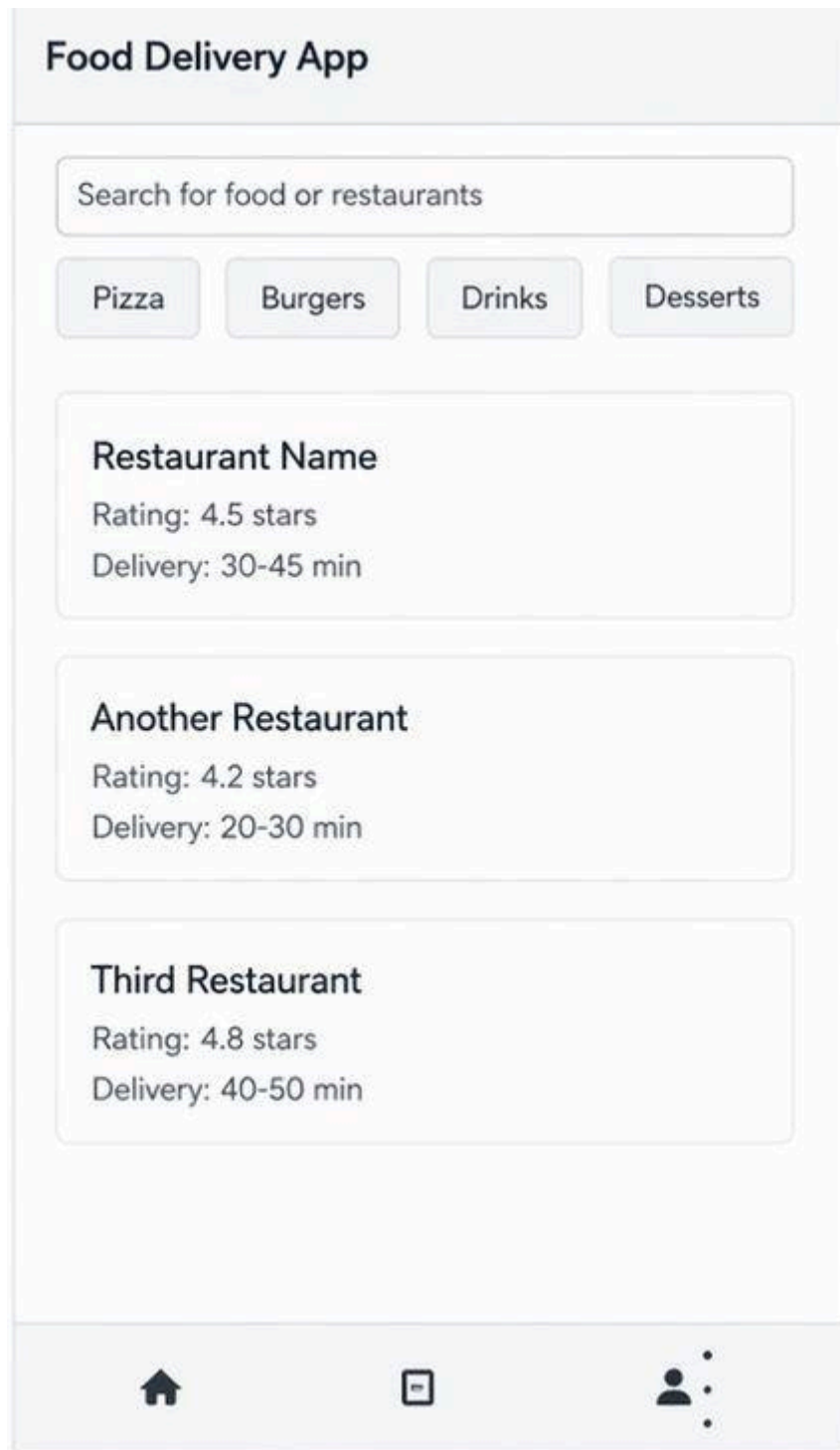
Track order status

2. User Design (UI/UX Design)

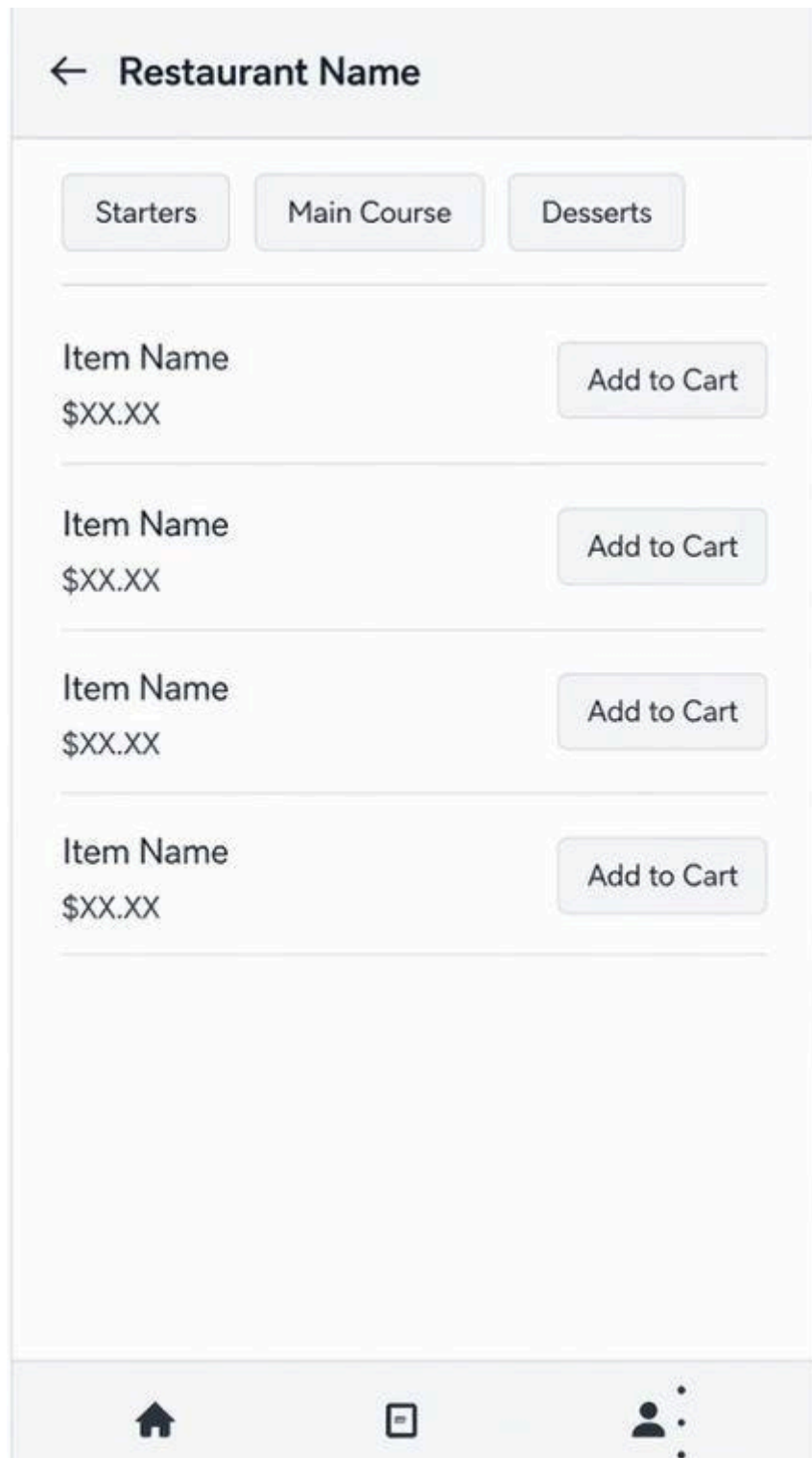
The interface was designed focusing on simplicity and smooth navigation.

Screens Designed:

- ◆ **Home Screen**



- Search bar to find food or restaurants
- Category buttons (Pizza, Burgers, Drinks, Desserts)
- List of restaurants with ratings and delivery time
- ♦ **Restaurant Menu Screen**



- Displays food categories (Starters, Main Course, Desserts)
- Shows food items with price
- “Add to Cart” button for each item

◆ **Cart Screen**

Your Cart

Item Name (x1)

\$XX.XX

Remove

Item Name (x1)

\$XX.XX

Remove

Total:

\$XXX.XX

Delivery Address

123 Main St, Anytown, USA

Place Order

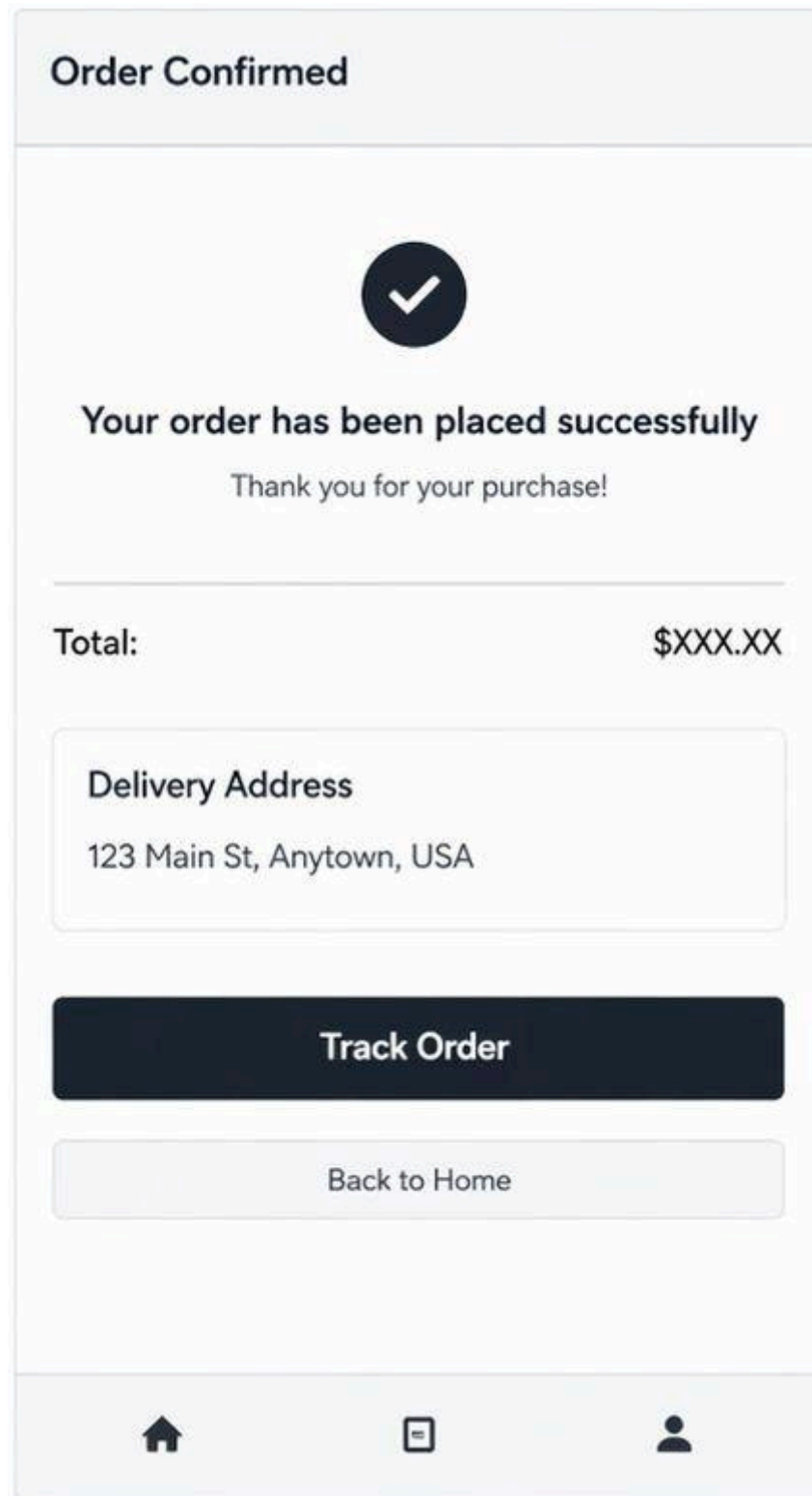






- Displays selected items in cart
- Option to remove items
- Shows total price
- Delivery address section

- “Place Order” button
- ◆ **Order Confirmation Screen**



- Displays success message

- Shows total amount
 - Displays delivery address
 - “Track Order” and “Back to Home” options
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3. Construction (Prototype Development)

The prototype was developed using Balsamiq.

Implementation Details:

Each screen was created as a separate wireframe

Buttons were connected to simulate navigation

Navigation Flow:

Home → Menu Screen

Add to Cart → Cart Screen

Place Order → Order Confirmation

Back to Home → Home Screen

Features Implemented:

Interactive buttons

Page navigation

Cart and order simulation

This stage shows fast UI creation and interaction as per RAD principles.

4. Testing and Evaluation

The prototype was tested using the preview feature in Axure.

Testing Results:

- All navigation flows worked correctly
- Buttons responded as expected
- User journey was smooth and logical

Limitations:

- No real backend integration
- Payment system is simulated
- Limited product filtering options

Future Improvements:

- Add cart management system
- Include multiple payment options
- Improve UI responsiveness
- Add user authentication



Result

A functional prototype of a food delivery mobile application was successfully designed using the RAD model. The project demonstrates all stages of UI development and provides a smooth user experience.